

CHAPTER 11

NUTRITION STRATEGIES FOR MAXIMIZING PERFORMANCE

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After completing this chapter, you will be able to

- list pre-, during-, and postcompetition nutrition recommendations for different sports;
- understand nutrition guidelines for aerobic exercise performance and recovery;
- understand nutrition recommendations for strength and hypertrophy;
- provide guidelines for weight gain and weight loss;
- identify hydration recommendations;
- recognize signs and symptoms of relative energy deficiency syndrome (RED-S) and eating disorders;
- understand the importance of having an intervention and referral system in place for athletes suspected of having an eating disorder;
- recognize the prevalence and etiologies of obesity; and
- assist in the assessment process for obese individuals.

What athletes eat and drink before and during competition can affect their performance, while their postevent meal has a greater impact on their recovery and, if the time period between events is less than 24 hours, performance in the next event or game. Therefore, this chapter focuses on pre-, during-, and postcompetition nutrition while also providing guidelines for athletes who want to lose or gain weight. Additionally, no talk about weight is complete unless it includes information on disordered eating and eating disorders. It is imperative that the strength and conditioning professional recognize when an athlete may

have signs and symptoms of an eating disorder and be an active member of the treatment team.

Precompetition, During-Event, and Postcompetition Nutrition

Athletes' dietary practices over time will influence their overall health and performance. Additionally, what athletes eat before and during competition can have both physiological and psychological effects on performance, while their postcompetition meal affects recovery and may therefore influence how they perform during their next bout of training or competition.

Precompetition Nutrition

The **precompetition meal** helps provide fluid to maintain adequate hydration, carbohydrate to maximize blood glucose and stored glycogen levels, and protein to reduce muscle soreness (7, 31). Glycogen is the main form of energy used during high-intensity ($>70\%$ $\dot{V}O_2\text{max}$) exercise; once these stores become depleted, the athlete will experience muscular fatigue (60). Small quantities of glycogen are stored in liver and muscle, totaling approximately 15 grams of glycogen per kilogram body weight (3). For example, an 80 kilogram (176 lb) man can store approximately 1,200 grams of glycogen. Glycogen stored in the liver is used for the whole body, while glycogen stored in muscle tissue is used by muscle (60).

Despite the vital roles that both hydration and glycogen play in athletic performance, studies examining the importance of the precompetition meal and its effect on performance are equivocal due to differences in study subjects and methods used. Some show that a high-carbohydrate preexercise meal enhances aerobic time to exhaustion (29, 98) as well as anaerobic performance in adolescent males (75), while no effect on time trial performance has been reported in the literature (105). Despite these differences and the fact that studies simulating performance cannot take into account several factors that distinguish the competitive environment from a lab setting, such as precompetition nerves, temperature, humidity, and altitude, athletes can adapt general precompetition guidelines based on the literature to fit their own unique needs and the competitive environment they are in. Importantly, an individualized precompetition meal should be consumed.

All precompetition meals should take timing, meal and fluid composition, event or sport, and individual athlete preferences into account. In order to minimize the potential for stomach upset, smaller quantities of fluids and food should be consumed when the precompetition meal is consumed closer in temporal proximity to competition. Precompetition foods and beverages should be familiar to the athlete (i.e., tried in practice), low in fat and fiber so that they

empty rapidly from the stomach and minimize any potential for gastrointestinal distress, and moderate in protein (protein promotes longer-lasting satiety and accelerates recovery) (7).

Athletes can choose from either high or low glycemic index (131) (see [chapter 10](#) for more details) carbohydrates before competition based on individual preference or tolerance and fiber content; research does not indicate that one is more advantageous than the other (15). Even though consuming carbohydrates that rapidly increase insulin, such as glucose, will lead to an initial drop in blood sugar at the beginning of exercise, blood sugar levels typically return to normal within approximately 20 minutes, and the initial drop has no negative effect on performance (82).

Minimizing Gastrointestinal Issues

To minimize the likelihood of stomach discomfort during competition, athletes should do the following:

- Try food in practice first. Always try new foods during several practice sessions before competition (7).
- When the meal is closer in time to the start of the game or event, consume smaller amounts of food and liquids.
- Avoid high-fat and high-fiber foods. Both fat and fiber slow down the rate of digestion. When food is still being digested as an athlete exercises, he or she may experience stomach cramps (7).
- Avoid sugar alcohols. Despite their name, sugar alcohols contain no alcohol, but their chemical structure resembles that of both sugar and alcohol. Sugar alcohols are a type of carbohydrate that is not completely absorbed in the gut. Consequently, consumption can cause gas, bloating, and cramping and may have a laxative effect. Sugar alcohols are found in some low-carbohydrate and sugar-free products, including sugar-free gum, toothpaste, and energy drinks. The two sugar alcohols most likely to cause gastrointestinal issues are sorbitol and mannitol (4). Any product that could result in consumption of 20 grams of mannitol must carry this warning on the label: "Excess consumption may have a laxative effect." Individual responses to sugar alcohol vary. Sugar alcohols include xylitol (the most common one found in dental products), erythritol, sorbitol, mannitol, maltitol, isomalt, lactitol, hydrogenated starch hydrolysates, and hydrogenated glucose syrups (4).

Aerobic Endurance Sports

The precompetition meal may be most important for aerobic endurance athletes who compete in long-duration activity (>2 h) in the morning after an overnight fast. Upon waking in the morning, blood sugar levels are low and liver glycogen stores are substantially reduced. Both conditions decrease the amount of carbohydrate available for use as energy. Carbohydrates at the precompetition meal can significantly enhance glycogen stores and improve exercise time to

exhaustion in those who regularly include carbohydrate in their diets, when consumed 3 or more hours before competition (33, 146).

A crossover study examined whether a precompetition high-carbohydrate meal combined with a carbohydrate-electrolyte sports drink consumed during exercise improved aerobic endurance running capacity (70% $\dot{V}O_2\text{max}$ until exhaustion) more than the sports drink alone (28). The scientists had men perform three treadmill runs, each separated by one week, after an overnight fast followed by a carbohydrate meal 3 hours before exercise in addition to a carbohydrate-electrolyte sports drink during running; a carbohydrate meal 3 hours before exercise and water during running; or a low-calorie placebo drink (same flavor as the sport drink) 3 hours before exercise and water during running. They replicated the same diet for the two-day period before each trial. No differences were noted in average daily caloric intake between the three trials. Consumption of the high-carbohydrate preexercise meal improved aerobic endurance running capacity by 9% compared to the placebo drink before exercise and water during the run. However, the high-carbohydrate meal plus sport drink during the run improved aerobic endurance capacity by 22% compared to the placebo beforehand and sport drink during the run. These results indicate that a high-carbohydrate preexercise or precompetition meal can help improve aerobic endurance running capacity (28).

Aerobic endurance athletes who are not chronically adapted to a low-carbohydrate diet and start exercise with depleted glycogen stores will break down muscle to use amino acids for energy and may acutely suppress immune and central nervous system functioning. Therefore, a high-carbohydrate preexercise meal can help attenuate the breakdown of skeletal muscle while also providing carbohydrate for immune and nervous system functioning (23, 76). Over time, adaptation to chronic consumption of a low-carbohydrate diet increases the body's reliance on its vast storage of fat as a fuel source during exercise, though training with low glycogen stores may suppress immune and central nervous system functioning (23, 53). In a study examining protein catabolism, six subjects rode a cycle ergometer for 1 hour at 61% $\dot{V}O_2\text{max}$ after a **carbohydrate loading** protocol or carbohydrate depletion. In the carbohydrate-depleted state, protein breakdown was calculated at 13.7 grams per hour, which represented 10.7% of the calories used during exercise (76).

It is clear that a preexercise meal consumed hours before competition can help improve aerobic endurance performance. However, extremely early morning start times can complicate the desire to feed with a concurrent desire for maximum sleep. For instance, a runner who must be at the starting line of a race at 7:00 a.m. may find that waking up at 3:00 or 4:00 a.m. to eat is not practical. Athletes who find themselves in this scenario should practice eating small

amounts of food 1 to 2 hours before they start, transitioning to easily digestible carbohydrates and liquid carbohydrates before competition, while also ensuring that they consume an adequate amount of carbohydrate during competition.

The following are general recommendations that can be adapted to meet each athlete's individual needs. More research needs to be done to examine the needs of athletes in a variety of sports and to determine the effects of varying amounts of each macronutrient on performance (100). Until this is done, the guidelines for aerobic endurance athletes are adapted to other sports. This information should also be applied to strength, power, and team or field sport athletes.

- Preexercise feeding may be more important than postexercise feeding if an athlete is underfueled. Female athletes may see greater benefits from their training when nutrients are consumed before exercise compared to after exercise on strength and lean mass adaptations (111).
- Athletes should prehydrate, if necessary, several hours before exercise to allow for fluid absorption and urine output. Their urine specific gravity (USG) reading should be <1.02 (120).
- Athletes who tend to get nauseated easily, who have experienced diarrhea during competition, who get anxious or experience precompetition jitters, or who compete in high-intensity sports (jarring movements can increase the likelihood of stomach upset), as well as those who are competing in the heat, may want to consider eating at least 4 hours before competition, as well as trying a variety of types of low-fiber carbohydrate options. Aerobic endurance athletes who eat at least 4 hours before competition should include approximately 1 to 4 grams of carbohydrate per kilogram body weight and 0.15 to 0.25 grams of protein per kilogram body weight (131).
- If the precompetition meal is consumed 2 hours before exercise, athletes should aim for approximately 1 gram of carbohydrate per kilogram body weight. Athletes should follow an individualized hydration plan. During prolonged activity in hot weather, they should consume a sport drink containing 20 to 30 mEq of sodium (460-690 mg with chloride as the anion) per liter, 2 to 5 mEq of potassium (78-195 mg) per liter, and carbohydrate in a concentration of 5% to 10% (120).
- When consumed closer to the start of competition, the preexercise meal should be smaller. Additionally, liquid sources of carbohydrate may be preferable when the preexercise meal is 1 hour before the event because they are emptied from the stomach faster than solid food (7). Gels, gummies, and similar sources of carbohydrate are also digested very quickly. [Table 11.1](#) summarizes these recommendations while providing sample food options.

TABLE 11.1 Precompetition Food and Fluid Recommendations for Aerobic Endurance Athletes

Time before competition	Food recommendations	Fluid recommendations	Sample precompetition meals for a 68 kg (150 lb) athlete	Sample meals based on food and fluid recommendations	
1 h	0.5 g carbohydrate per kilogram body weight		34 g carbohydrate	• 1 small banana • 8 fl oz (237 mL) sport drink	• 37 g carbohydrate • 8 fl oz (237 mL) fluid
2 h	1 g carbohydrate per kilogram body weight	If not adequately hydrated, sip on 3-5 mL (0.10-0.17 fl oz) of fluid per kilogram body weight (120)	68 g carbohydrate 7-12 fl oz (207-355 mL) fluid	• 2.5 cups (525 g) boiled, plain potato, flesh only • 2 mini bagels with 1 tbsp (20 g) jam • 8 fl oz (237 mL) sport drink	• 66 g carbohydrate • 72 g carbohydrate • 8 g protein • 8 fl oz (237 mL) fluid
4 h or more	1-4 g carbohydrate per kilogram body weight 0.15-0.25 g protein per kilogram body weight (105)	Should consume approximately 5-7 mL (0.17-0.24 fl oz) of water or a sport drink per kilogram body weight (120)	68-273 g carbohydrate 10-17 g protein Minimal fat	• Bowl of cereal and fruit (8 fl oz [237 mL] skim milk, 2 cups [55 g] Cheerios, 1/4 cup [40 g] unpacked raisins)	• 74 g carbohydrate • 11 g protein • 14 fl oz (414 mL) water
				• Egg white sandwich (2 egg whites, 2 pieces of	• 72 g carbohydrate (based on 1 slice = 64 g weight) • 17 g protein

Note: These recommendations can be adapted for other types of athletes.

Time before competition	Food recommendations	Fluid recommendations	Sample precompetition meals for a 68 kg (150 lb) athlete	Sample meals based on food and fluid recommendations	
				white bread) (skip high-fiber breads before workout)	14 fl oz (414 mL) water

Note: These recommendations can be adapted for other types of athletes.

Athletes may want to keep a record of their food intake, including the time each meal or snack was consumed and how they felt while training. By recording the types and amounts of food consumed and when these are consumed in relation to training, athletes may be able to identify any issues they are having with performance or stomach upset and develop a better precompetition plan. Carbohydrate foods that are high in fructose, sorbitol, and fermentable oligosaccharides, disaccharides, monosaccharides, and polyols (FODMAPs) may increase the risk of gastrointestinal distress when consumed before exercise. Selecting foods that are lower FODMAP and lower fructose may minimize the likelihood of gastrointestinal issues.

The primary purpose of the precompetition meal is to provide sufficient fluid to maintain hydration and carbohydrate to maximize blood glucose and stored glycogen while also satisfying hunger.

Carbohydrate Loading

Depletion of muscle and liver glycogen leads to fatigue during long-term aerobic endurance exercise (117, 129, 141). Therefore, a technique called *carbohydrate loading* has been used for decades to enhance muscle glycogen before aerobic endurance events. Although there are many different variations of carbohydrate loading, all include high carbohydrate intake in the days leading up to an event to maximize glycogen stores and therefore carbohydrate availability in the later stages of the event (32). Carbohydrate loading offers potential benefits for athletes training and competing for more than 90 minutes, such as distance runners, road cyclists, cross-country skiers, and other aerobic endurance athletes who risk depleting glycogen stores, and it may conceivably benefit other athletes as well (112).

A commonly used carbohydrate loading regimen includes three days of a high-carbohydrate diet in concert with tapering exercise the week before competition and complete rest the day before the event. The diet should provide adequate calories and carbohydrate per day: 8 to 10 grams of carbohydrate per kilogram body weight. This regimen should increase muscle glycogen stores 20% to 40% above normal (32). However, higher intakes, 10 to 12 grams of carbohydrate per kilogram body weight, have been suggested for runners during the 36 to 48 hours before a marathon (22).

Studies show that carbohydrate loading is effective in men (129). However, the studies in women have been mixed. One study examined the effects of carbohydrate loading on aerobic endurance performance and substrate use in eight 20- to 40-year-old weight-stable, eumenorrheic female runners with an average running history of 53 kilometers (33 miles) per week for at least 12 months before the study and a typical carbohydrate intake below 65% of total caloric intake (9). Each female completed three different 24.2-kilometer (15-mile) self-paced treadmill performance runs after four days on each experimental diet regimen: carbohydrate supplementation (50% of calories from carbohydrate), carbohydrate loading and supplementation (75% of calories from carbohydrate), and placebo (50% of calories from carbohydrate). Both the carbohydrate-supplemented and carbohydrate loading plus supplementation groups consumed a 6% carbohydrate-electrolyte solution before exercise (6 mL/kg of body weight) and every 20 minutes during exercise (3 mL/kg of body weight). And though the carbohydrate-supplemented and carbohydrate loading plus supplementation groups used a greater portion of energy from carbohydrate during their runs, there was no significant difference in running performance between the groups. However, both total calories and grams of carbohydrate per day were not reported; therefore, it is possible that total calorie or carbohydrate intakes (or both) were not adequate for the runners. Additionally, given the small number of subjects included in this study, differences in performance may have been noted if more subjects had been included (9).

In another study, researchers found that males increased glycogen content significantly while female athletes did not after increasing carbohydrate intake from 58% to 74% of calories for four days before a submaximal aerobic endurance exercise test, due to either inadequate carbohydrate or calorie intake (or both) or sex differences in glycogen storage (129). In a follow-up study, the same researchers examined glycogen storage capacity in six well-trained men and six well-trained women (132). Subjects were randomly assigned to one of three diets for four days: a high-carbohydrate diet (75% of total calorie intake), a high-carbohydrate diet plus extra calories (75% of calories from carbohydrate and a 34% increase in total calories), or their habitual daily diet. In men, both

the higher-carbohydrate diet and higher-carbohydrate plus extra calories diet led to significantly greater levels of glycogen than the habitual diet. However, in women, only the higher-carbohydrate plus extra calories diet resulted in significant increases in glycogen storage compared to the habitual diet. In the high-carbohydrate conditions, men increased their total carbohydrate intake to 7.9 grams of carbohydrate per kilogram body weight, whereas women reached just 6.4 grams of carbohydrate per kilogram body weight. No differences were noted between men and women in their ability to use glycogen as measured by enzymatic activity. Therefore, the failure of previous studies to show improved performance from carbohydrate loading in women was likely due to inadequate overall carbohydrate intake (132). This theory (that women were not consuming enough total carbohydrate in many of the protocols) was backed up in a study examining female cyclists. In trained female cyclists, three or four days on a moderate-carbohydrate diet (48% of calories from carbohydrates) followed by three or four days consuming 78% of calories from carbohydrates (8.14 g of carbohydrate per kilogram body weight) led to significant increases in glycogen stores and greater cycling time to exhaustion compared to seven days on the moderate-carbohydrate diet (142). Researchers have found that glycogen storage capacity in the 4-hour time period after exercise did not differ between men and women when they consumed the same amount of carbohydrate per unit body weight (1 g of carbohydrate per kilogram body weight immediately and 1 hour after the completion of exercise) (130). And while glycogen storage capacity is greater during the luteal phase of the menstrual cycle in women as compared to the early follicular phase due to hormonal differences, carbohydrate loading and increased carbohydrate intake can make up for this difference (99).

The main challenge with carbohydrate loading in females appears to be their overall daily calorie intake. Women who habitually consume less than 2,400 calories per day may find it difficult to consume greater amounts of carbohydrate. Therefore, female athletes may need to increase their total energy intake above 2,400 calories, in addition to consuming a higher-carbohydrate diet, in order to increase glycogen stores (129, 142).

Though the majority of studies on carbohydrate loading have been done in aerobic endurance athletes, some research has attempted to examine carbohydrate loading for athletes in high-intensity sports as well. In a randomized, crossover design, seven professional soccer players who consumed an average of 46% of calories from carbohydrates ate either a 39% or a 65% carbohydrate diet two days before being testing. Each test consisted of 6,858 meters (7,500 yd) of field work (completed at 65%, 57%, and 81% $\dot{V}O_{2\max}$) followed by a treadmill run to exhaustion in an attempt to mimic a soccer game. After the high-carbohydrate diet, the players ran 17.1 kilometers (10.6 miles)

total, which was 0.9 kilometers (0.6 miles) longer (a significant difference) than the distance run while on the lower-carbohydrate diet. However, three of the athletes ran less than 420 meters (459 yd) longer, indicating significant variability in the athletes' responses to a higher-carbohydrate regimen (14). The results of this study may or may not be applicable to soccer players, since the average distance covered in a soccer match is considerably shorter (10.3 km [6.4 miles] with a range of 9.7 to 11.3 km [6.0-7.0 miles], as reported in one study), and it is covered in an intermittent manner and at varying levels of intensity (72).

In a study examining the effect of carbohydrate loading on resistance exercise performance, eight healthy young men were randomly assigned to either a higher-carbohydrate (6.5 g of carbohydrate per kilogram body weight) or a moderate-carbohydrate diet (4.4 g of carbohydrate per kilogram body weight) for four days. After this period, they participated in a resistance exercise test including 4 sets of 12 repetitions of maximal-effort jump squats with a load of 30% of a 1-repetition maximum (1RM) and a 2-minute rest period between sets (52). There was no significant difference in power performance between subjects on the high-carbohydrate and the moderate-carbohydrate diets. Therefore, in this study a higher-carbohydrate diet did not enhance power performance during a 4-set resistance exercise test. However, it is unclear whether the carbohydrate loading regimen could have affected power performance if more sets were performed or if the carbohydrate loading regimen more closely resembled that of an aerobic endurance athlete, providing 8 to 10 grams carbohydrate per kilogram body weight (52).

The degree of benefit derived from carbohydrate loading varies between individuals, even between aerobic endurance athletes, and therefore athletes should determine the value of this regimen while weighing any negative side effects, such as temporary weight gain, in practice before competition. Athletes who use carbohydrate loading as a means to maximize glycogen storage should also know how different types of carbohydrates affect them. For instance, oligosaccharides, which are found in dry beans and peas, onions, and foods with added inulin or other oligosaccharides (such as some nutrition bars and shakes), are rapidly fermented by bacteria in the gut, which can lead to excessive intestinal gas and bloating. Dietary fiber is found in vegetables, fruits, whole grains, dry beans and peas, nuts, and seeds (140). [Table 11.2](#) provides a sample daily meal plan for a 68-kilogram (150-pound) athlete following research-based carbohydrate recommendations for aerobic endurance athletes.

Carbohydrate loading is an effective strategy to maximize glycogen storage. However, athletes must consume 8 to 10 grams of

carbohydrate per kilogram body weight per day during the loading period to notice any benefit from carbohydrate loading.

During-Event Nutrition

Nutrition is an important factor during aerobic endurance events lasting greater than 45 minutes, intermittent-activity sports, or when an athlete has multiple events in one day. Fluids and carbohydrates can directly affect performance. Amino acid oxidation, primarily the branched chain amino acids, are oxidized during sustained aerobic endurance exercise, providing substrate for gluconeogenesis. Easily digestible proteins can minimize muscle damage and substrate depletion (116).

Proper hydration during competition is essential for performance while also helping to prevent overheating, dehydration, and heat illness. Athletes should hydrate several hours before exercise to allow for fluid absorption and urine output before competing. Additionally, they should consume enough fluid during exercise to prevent water weight losses exceeding 2% of body weight (120). The optimal sport drink contains 20 to 30 mEq of sodium (460-690 mg with chloride as the anion) per liter, 2 to 5 mEq of potassium (78-195 mg) per liter, and carbohydrate in a concentration of 5% to 10% (2). Sport drinks with a higher concentration of carbohydrate—those that contain more than 8%—delay gastric emptying (how quickly the drink is emptied from the stomach), which could lead to stomach discomfort (87). Therefore, a carbohydrate concentration of 6% to 8% may be ideal (120).

TABLE 11.2 Carbohydrate Loading Regimens and Sample Daily Meal Plans for Aerobic Endurance Athletes

Sample meal plan for a 68 kg (150 lb) athlete		
Amount of carbohydrate		Nutrition breakdown
8-10 g carbohydrate per kilogram body weight consumed on day 3 before an event (30)	Breakfast: • Turkish menemen (3 large whole eggs [150 g], 1 ripe tomato, 1/2 green bell pepper) • 4 slices (240 g) multigrain toast • 1 cup (150 g) mixed berries • 5.3 oz (150 g) plain Greek yogurt Snack: • 6 tbsp (90 g) hummus • 1 whole wheat pita bread (60 g) Lunch:	~8.1 g carbohydrate per kilogram body weight (i.e., 550 g total carbohydrate) 100 g fat 175 g protein

Note: These recommendations can be adapted for other types of athletes. Macronutrient calculations are approximate.

Sample meal plan for a 68 kg (150 lb) athlete		
Amount of carbohydrate		Nutrition breakdown
	- Chicken stir fry (2 cups [320 g] brown rice, 6 oz [170 g] cooked chicken breast, 1 cup [150 g] mixed stir fry vegetables cooked in 2 tbsp [30 mL] sesame oil) Early afternoon snack: 1 bagel (100 g) with 2 tbsp (40 g) 100% fruit jam Post-taper run: 8 fl oz (237 mL) 100% juice 1 cup (150 g) granola 5.3 oz (150 g) Greek yogurt Dinner: - Indian chicken tikka masala (7 oz [200 g] chicken breast, 6.8 fl oz [200 mL] tikka masala sauce, 5.3 oz [150 g] cooked basmati rice, 1.8 oz [51 g] steamed broccoli)	
10-12 g carbohydrate per kilogram body weight consumed on each of the two days before an event (22)	Breakfast: - Turkish menemen (3 large whole eggs [150 g], 1 ripe tomato, 1/2 green bell pepper) - Swedish overnight oats (2.5 oz [70 g] rolled oats, 5.3 oz [150 g] Greek yogurt, 5.3 oz [150 g] mixed berries, 1 tbsp [21 g] honey) Snack: 8 tbsp (113 g) hummus 2 whole wheat pita bread (120 g) Lunch: - Chicken stir fry (2 cups [320 g] brown rice, 4 oz [113 g] cooked chicken breast, 1 cup [150 g] mixed stir fry vegetables, all cooked in 2 tbsp [30 mL] sesame oil) Early afternoon snack: 1 bagel (100 g) with 2 tbsp (40 g) 100% fruit jam Post-taper run: 8 fl oz (237 mL) 100% juice 1 packet (43 g) instant oatmeal 1/3 cup (50 g) packed raisins added 8 fl oz (237 mL) skim milk Dinner: - Indian chicken tikka masala (7 oz [200 g] chicken breast, 6.8 fl oz [200 mL] tikka masala sauce, 5.3 oz [150 g]	~10.3 g carbohydrate per kilogram body weight (i.e., 700 g total carbohydrate) 105 g fat 210 g protein

Note: These recommendations can be adapted for other types of athletes. Macronutrient calculations are approximate.

Sample meal plan for a 68 kg (150 lb) athlete		
Amount of carbohydrate		Nutrition breakdown
	cooked basmati rice, 1.8 oz [51 g] steamed broccoli	

Note: These recommendations can be adapted for other types of athletes. Macronutrient calculations are approximate.

Fluid intake guidelines are different for children. According to the American Academy of Pediatrics, children weighing 40 kg (88 pounds) should drink 5 fluid ounces (148 mL) of cold water or a flavored salted beverage every 20 minutes during practice, while adolescents weighing 60 kg (132 pounds) should drink 9 fluid ounces (266 mL) every 20 minutes even if they do not feel thirsty. They also recommend a sodium chloride concentration of 15 to 20 mmol/L (approximately 1 g per liter), which has been shown to increase voluntary hydration by 90% when compared to unflavored water (30, 106, 147). Despite these recommendations, keep in mind that 9 ounces of fluid at one time is a considerable amount for an adolescent and may cause stomach distress. Therefore, these guidelines may need to be adjusted for individual athletes, depending on size and sweat rates.

Aerobic Endurance Sports

Consuming carbohydrates during prolonged aerobic endurance exercise can improve performance while also reducing exercise-induced stress and suppression of immune system functioning (95). And though sport drinks provide carbohydrates, they cannot keep up with an athlete's carbohydrate use during prolonged, intense activity (unless an excessive amount of fluid is consumed). In fact, athletes who train intensely may burn 600 to 1,200 calories or more per hour (70). Several studies show that a carbohydrate intake ranging from 28 to 144 grams per hour (with higher amounts studied during cycling) during an aerobic endurance activity can decrease reliance on limited glycogen stores, extend time to exhaustion, and improve performance by providing a steady stream of carbohydrate that can be used for energy (32, 66, 137).

Despite greater intakes per hour, as noted earlier, **exogenous** (external; that which is consumed) carbohydrate oxidation rates do not exceed 1.0 to 1.1 gram per minute, likely due to the rate of glucose absorption and possibly also the limited delivery rate of 1 gram per minute of glucose into the bloodstream (65). However, each carbohydrate has a different rate of oxidation. Glucose, sucrose, maltose, maltodextrins, and amylopectin are oxidized quickly; the oxidation rates of fructose, galactose, and amylose are 25% to 50% slower (65). Additionally, each type of carbohydrate has a different intestinal transport system. If an athlete consumes one type of carbohydrate—fructose, for instance—

carbohydrate digestion will be limited when the intestinal transporter for fructose becomes saturated. Therefore, consuming multiple types of carbohydrates together, such as sucrose, fructose, and glucose or maltodextrin, increases the rate of carbohydrate absorption and exogenous carbohydrate oxidation compared to consuming an **isocaloric** amount of just one type of sugar (62). In addition to improving the rate of carbohydrate use by the body, consuming multiple types of carbohydrate during exercise was found to improve cycling time-trial performance after 120 minutes of cycling (the study design simulated the last few stages of an aerobic endurance race when an athlete must provide maximum effort) when subjects were provided glucose combined with fructose versus glucose only at a rate of 1.8 grams per minute (108 g of carbohydrate per hour) (34). Additionally, athletes given 36 grams of glucose and fructose every 15 minutes had a better 100 kilometer (62 miles) cycling time than cyclists given only glucose every 15 minutes (137).

In addition to actually ingesting carbohydrates, simply rinsing carbohydrates through the mouth (without actually ingesting them) seems to improve performance lasting approximately 1 hour by 2% to 3%, presumably by affecting the central nervous system (64).

In addition to providing carbohydrates during an aerobic endurance activity, adding protein to a carbohydrate gel led to increased time until exhaustion during a bout of cycling while also attenuating the rise in **creatine kinase** (a marker of muscle damage) during cycling, due either to the protein itself or to the additional calories provided by the protein (119). A meta-analysis and review of the research examining how the addition of protein to carbohydrate affects aerobic endurance performance revealed mixed findings (119). Only time-to-exhaustion studies showed that the addition of protein made a significant difference. However, these studies were not controlled for calorie intake during the trial; therefore, it is unclear if any benefit was due to protein or to the added calories provided by the protein. The three time-trial studies showed no difference between carbohydrate only and carbohydrate plus protein (127).

Fructose and Gastrointestinal Symptoms

Fructose, a sugar found naturally in fruit and also added to a number of foods and beverages, including many sports nutrition products, is often blamed for symptoms of stomach upset during exercise. Though some people do not completely absorb fructose and may therefore experience bloating, gas, abdominal discomfort, and alterations in bowel functioning after consuming this sugar (123), studies have not examined gastrointestinal symptoms related to fructose ingestion alone in athletes. Thus, athletes need to test, in practice, whether changing their sport drinks or other products they consume during training (and testing fructose-free conditions) helps alleviate any gastrointestinal symptoms they may be having (126). Athletes with irritable bowel syndrome are more likely to have a problem absorbing fructose and therefore experience side effects (123).

Intermittent High-Intensity Sports

Many team sports, such as soccer, tennis, basketball, and American football, include repeated bouts of short-duration, high-intensity activity in addition to involving a wide range of skills. Fatigue during play could stem from a number of factors, including decreased or depleted glycogen stores and dehydration. The provision of both fluids and carbohydrate is essential for performance during prolonged intermittent sports. For instance, a long tennis match may last 4 hours—considerably longer than many aerobic endurance events. Because tennis players cannot drink continuously during a match and can lose more than 2.5 liters (84.5 fl oz) of fluid per hour, a recommendation of 200 to 400 milliliters (6.7-13.5 fl oz) per changeover (switching sides) has been put forth for these athletes (69). In addition to maintaining hydration status, research shows that carbohydrate supplementation may not affect ratings of perceived exertion (46) but improves stroke quality (which includes measures of velocity, precision, and error rate) during prolonged play (141).

In soccer players, 5 milliliters (0.2 fl oz) per kilogram body weight of a 6.9% glucose-polymer drink, consumed 15 minutes before each match and at halftime, made no difference in several measures of performance such as successful tackles, heading, dribbling, or shooting ability (150). Yet another soccer study, this one with professional soccer players, found that the group consuming a carbohydrate-electrolyte beverage improved some parameters of performance during match play. In this study, 22 professional male soccer players consumed the same diet for seven days (55% of the calories were derived from carbohydrate, 25% from fat, and 20% from protein), refrained from exercise for three days before the match, and consumed a standardized breakfast 4 hours before the match on game day. They were divided into two groups, receiving either a carbohydrate-electrolyte drink (7% carbohydrates, sodium 24 mmol/L, chloride 12 mmol/L, potassium 3 mmol/L) or placebo. Each group drank 5 milliliters (0.2 fl oz) per kilogram body weight before the match and 2 milliliters (0.1 fl oz) per kilogram body weight every 15 minutes during the 90-minute game. The group receiving the carbohydrate-electrolyte drink finished the specific dribble test faster than those receiving the placebo. In addition, ratings of precision were higher in the carbohydrate-electrolyte trial versus placebo. However, there were no differences in the coordination or power tests between groups. In this particular study, supplementation with a carbohydrate-electrolyte drink improved soccer-specific skill performance compared with a placebo (101).

In another study that found that during-play carbohydrate consumption was beneficial, a randomized, double-blind protocol was used. Seventeen male soccer players received 8 milliliters (0.3 fl oz) of a 6.4% carbohydrate-electrolyte beverage per kilogram body weight before exercise plus 3 milliliters (0.1 fl oz)

per kilogram body weight after every 15 minutes (for a total of 52 g of carbohydrate per hour) during a 90-minute intermittent shuttle test (6). Consuming the carbohydrate-electrolyte beverage attenuated a significant decrease in skill reduction from before exercise until the last 15 to 30 minutes of exercise compared to placebo. In the carbohydrate trial, there was a 3% reduction in skill performance during this time period, while in the placebo trial a 15% decline in skill performance was noted. However, this trial was performed after carbohydrate-depleting exercise followed by a low-carbohydrate meal and then an overnight fast; therefore, the effect of carbohydrates on performance may depend not only on the amount consumed during intermittent activities but also on whether the athlete is competing in a fed or fasted and glycogen-depleted state (6).

In another study, which used four 15-minute quarters of intermittent high-intensity shuttles at various intensities (walking, jogging, running, sprinting, and jumping), separated by a 20-minute rest at halftime and followed by a shuttle run to fatigue (designed to mimic a competitive soccer or basketball game), subjects who consumed a **carbohydrate-electrolyte (carbohydrate) drink** before exercise (5 mL of a 6% solution per kilogram body weight) and at halftime (5 mL of a 18% solution per kilogram body weight) ran 37% longer in the run to fatigue than those receiving a placebo and were significantly faster in a 20 meter (22 yd) sprint during the fourth quarter. In addition, the carbohydrate-supplemented group performed better on a total body motor skill test during the later stages of exercise and reported decreased perceptions of fatigue, indicating that the consumption of a carbohydrate-electrolyte beverage during intermittent sports is advantageous (145).

Strength and Power Sports

Carbohydrates are also an essential source of energy used during resistance training and therefore in strength and power sports. Research studies using a series of different weightlifting protocols found that participants used a significant amount of muscle glycogen (78), though this is notably lower than what is typically seen for power–endurance or intermittent high-intensity sports. Most of these studies had participants perform just a few sets of exercises. Therefore, athletes who perform in strength and power competitions or sports, or in positions that rely on muscle strength and power (e.g., hammer throwers and offensive linemen in American football), could deplete their muscle glycogen stores to a level that affects performance. Furthermore, starting with already low carbohydrate stores will increase muscle breakdown (76). Strength and power athletes can maintain their glycogen stores, which may decrease muscular

fatigue in slow-twitch fibers and possibly lead to better performance, by supplementing with carbohydrate before and during competition (47).

During-Competition Food and Fluid Recommendations

Due to large variations in sweat rates and electrolyte concentrations, athletes should measure weight changes during training and competition in specific weather conditions and develop individualized hydration strategies based on this information (120). During prolonged activity in hot weather, the Institute of Medicine recommends sport drinks containing 20 to 30 mEq of sodium (460-690 mg with chloride as the anion) per liter, 2 to 5 mEq of potassium (78-195 mg) per liter, and carbohydrate in a concentration of 5% to 10% (2).

Hydration guidance for children and adolescents consists of 5 to 9 fluid ounces (148-266 mL) of cold water or a flavored salted beverage or sport drink depending on their weight. For example, children or adolescents weighing 40 kilograms (88 lb) should drink 5 fluid ounces (148 mL) and those weighing 60 kg (132 lb) should drink 9 fluid ounces (266 mL) every 20 minutes during practice, even if they do not feel thirsty (30, 37, 106). Aerobic endurance athletes should consume 28 to 144 grams of multiple types of carbohydrates together, such as sucrose, fructose, glucose, or maltodextrin, each hour during prolonged aerobic endurance activity (32, 34, 66, 137).

Postcompetition Nutrition

The postcompetition meal helps athletes rehydrate, replenish glycogen stores, and repair muscle tissue. Therefore, what they consume in the time period soon after training or competition helps optimize adaptations as well as prepare their body for the next bout of activity. This time frame has been referred to as a *garage door of opportunity* because the athlete's body is primed to assimilate nutrients and also because this previously assumed **anabolic window** is actually much longer than was originally thought (11). Each athlete's postcompetition needs vary based on the sport he or she plays, intensity during play, amount of time played, the individual's weight and age, and probably the person's sex. However, much more research has been conducted with male subjects as opposed to females; therefore, recommendations are not broken out based on sex due to a lack of sufficient data in females.

After competition, athletes should replace fluid and electrolyte losses. If time permits, normal meals, snacks (provided they contain some sodium), and water will restore fluid and electrolyte losses. More salt can be added to foods when sweat sodium losses are substantial (1, 120). Athletes can choose a carbohydrate-electrolyte sport drink or plain water alongside foods that contain sodium chloride (or they can salt their foods) since sodium is essential for helping the body retain fluid (83, 120, 125). Rehydration strategies should be individualized as much as possible. Athletes competing in weight-class sports such as wrestling and mixed martial arts may deliberately dehydrate themselves

to make weight and attempt to rehydrate before competition, though short time periods between weighing and competing may mean starting competition in a dehydrated state, leading to poor performance and health risks.

Aerobic Endurance Events

After prolonged aerobic endurance events, it is important to replenish carbohydrate stores before the next training session or competition (whichever comes first) and to aid in immune system function and consume enough protein to build and repair muscle. Glycogen synthesis occurs in two distinct phases. The first phase is independent of insulin and lasts 30 to 60 minutes, and glycogen synthesis occurs rapidly. The second phase lasts several hours, and glycogen synthesis occurs at a much slower pace. Glycogen synthesis occurs at a rapid rate when large amounts of carbohydrate, 1.0 to 1.85 grams per kilogram body weight per hour, are consumed immediately after exercise or competition and at regular intervals every 15 to 60 minutes thereafter for up to 5 hours (61). For some athletes, though, it may be difficult to ingest carbohydrate quantities at the upper end of that range. Though athletes may be able to fully replenish their glycogen stores right away or over the course of a 24-hour period after a long training session, this is not necessarily the case after competition. Strenuous aerobic endurance events that result in measurable muscle damage, such as the marathon, lead to delayed glycogen resynthesis even if an athlete consumes a higher-carbohydrate diet, possibly due to either metabolic disturbance or mechanical damage to muscle cells (12, 124).

Though athletes are often told that they need to eat carbohydrate immediately after they finish competing, research shows that this is not always necessary; they may be able to wait 2 hours after finishing a glycogen-depleting event before eating carbohydrate, although this may not necessarily be optimal. In a study examining the rate of glycogen resynthesis, five high-glycemic meals were given to athletes over the course of a 24-hour period after a 2-hour-long glycogen-depleting bout of cycling. One group received the first three meals over the 4-hour period after exercise, while the second group received the first three meals at regular 2-hour intervals starting 2 hours after they finished over the first 6 hours after exercise. The rate of glycogen resynthesis was the same between groups after 8 hours and after 24 hours (103). Therefore, athletes who have more than 24 hours to recover can likely wait before eating after exercise and replace their glycogen over the 24-hour time period after exercise as long as they consume an adequate amount of carbohydrate, though they may not receive the same systemic recovery benefits beyond glycogen replenishment. However, athletes who train two or three times a day or have less than 24 hours to recover may want to consider eating or drinking a high-carbohydrate meal

immediately after finishing their event and at regular intervals thereafter to quickly replenish glycogen stores.

Because prolonged aerobic endurance exercise breaks down muscle tissue, protein should be included in the aerobic endurance athlete's posttraining meal to help start muscle building and repair, which may attenuate posttraining and postevent muscle soreness (85). Consuming protein after training has another benefit: It increases the rate of glycogen storage if carbohydrate intake is less than 1.2 grams of carbohydrate per kilogram body weight per hour (61). In a randomized, controlled trial, 18 elite orienteers participated in 13 exercise sessions during one week. Half of the orienteers (the PRO-CON group) ingested a protein drink before (0.3 g per kilogram body weight) and a protein-carbohydrate drink (0.3 g protein per kilogram body weight + 1 g carbohydrate per kilogram body weight) after training. The other half of participants consumed an isocaloric carbohydrate-only beverage before and after exercise (CHO). Diet was kept consistent throughout the study (15%/63%/22% protein/carbohydrate/fat) except for the addition of the supplements. The basic diet plus supplements provided 3 grams of protein per kilogram body weight per day and 8.3 to 9.3 grams of carbohydrate per kilogram body weight per day in the PRO-CON group; the CHO group consumed 1.8 grams of protein per kilogram body weight per day and 8.8 to 10.8 grams carbohydrate per kilogram body weight per day. A 4-kilometer (2.5 miles) run test was performed at the beginning and end of the study and 2 hours after participants consumed a standardized breakfast. PRO-CON significantly improved performance and reduced markers of muscle breakdown, while CHO did not improve performance. However, it is not clear from this study if the timing of protein made the difference or the increased total daily intake of protein in the PRO-CON as compared to the CHO group (48).

Other research does suggest that protein is important after aerobic endurance exercise. For instance, a single-blind, randomized, triple-crossover design was used to assess how different doses of a protein-leucine blend affected myofibrillar protein fractional synthetic rate (FSR; muscle protein synthesis) in 12 aerobic endurance-trained men after 100 minutes of high-intensity cycling (performed 3 hours after a standardized breakfast; diet the day before the test was standardized based on calorie needs and provided the same percentage of carbohydrate, protein, and fat for each subject). Cyclists consumed 70 g/15 g/180 g/30 g protein/leucine/ carbohydrate/fat, or 23 g/5 g/180 g/30 g, or 0 g/0 g/274 g/ 30 g (control) in four servings during the first 90 minutes of a 240-minute (4-hour) recovery period. The lower-protein, lower-leucine supplement (23 g/5 g/180 g/30 g) increased FSR by 33% $\pm$ 12% compared to the control group, while additional protein and leucine (70 g/15 g/180 g/30 g) increased FSR

by $51\% \pm 12\%$ compared to the control group. Though no significant differences were noted between the two protein and leucine groups, both fared better than the placebo group, indicating that protein is important after a long, tough workout (118). Given different workouts, exercise completed in a fasted or unfasted state, and different doses of protein (10-96 g) used in other studies, the precise minimum dose of protein needed to maximize FSR after aerobic endurance exercise is not clear (21, 57). Additionally, it is not clear if maximizing FSR in the time period immediately after aerobic endurance exercise will translate to improved performance over time. And finally, the ideal time period after aerobic endurance training during which protein should be consumed is not clear and may depend on whether training was completed in a fasted or unfasted state as well as on total daily protein intake. However, one study found that delaying protein intake after aerobic endurance exercise by as little as 3 hours blunts its anabolic effects (77).

High-Intensity Intermittent Sports

Because athletes who participate in intermittent high-intensity sports—such as basketball, hockey, and soccer—may play more than one game per day, sometimes with just a few hours in between tournament games, immediate recovery after a game is imperative for performance in their next game. And when competing in sports that involve prolonged high-intensity intermittent activity, such as soccer, American football, field and ice hockey, rugby, and tennis, athletes may significantly reduce their muscle glycogen stores, leading to muscular and cognitive fatigue (13). Fully replacing muscle glycogen before a subsequent bout of exercise or competition may prolong time until fatigue and improve performance. In a study designed to mimic a soccer game, Nicholas and colleagues had six males complete the same exercise test separated by 22 hours on two consecutive days—75 minutes of a prolonged, intermittent, high-intensity shuttle run test after which they completed as many 20 meter (22 yd) shuttles as possible (each shuttle alternated between jogging and sprinting). Subjects were assigned to one of two diets for the 22-hour recovery period: a recovery diet including 10 grams of carbohydrate per kilogram body weight or an isocaloric diet with more protein and fat than their normal daily diet. Intermittent running capacity improved after the higher-carbohydrate diet as compared to the higher-calorie condition with more protein and fat (94). Also, Balsom and colleagues found that male study subjects performed significantly more work in both short-term (<10 min) and prolonged (>30 min) high-intensity intermittent exercise sessions after consuming a high- versus low-carbohydrate diet for the 48-hour period before the exercise sessions (13).

High-intensity intermittent sports can lead to some degree of muscle damage depending on many factors including the time and intensity of play and body size. Studies show that consuming protein after exercise helps decrease some markers of muscle damage (42). And though postexercise carbohydrate intake can affect performance in a subsequent event or training session, particularly when sessions are close together in time, the addition of protein will likely not affect performance in a subsequent bout of exercise during the 4-hour recovery period following the initial event or exercise session (85).

Strength and Power Sports

Athletes in strength and power sports rely on both blood glucose and glycogen for energy during competition. Because a single bout of resistance training can lead to significant reductions in glycogen (78) and decreases in glycogen can impair force production and isometric strength while accentuating muscle weakness, it is imperative that these athletes restore glycogen levels before their next bout of exercise (47). During the recovery period after strength and power competitions, athletes should focus on consuming higher-glycemic carbohydrates immediately after exercise if they must compete or train again over the course of the 24-hour period after the initial training session or competition. The amount of carbohydrate they need to fully replenish glycogen stores depends on many factors, including intensity and the time spent competing, their overall body weight and muscle mass, their preexercise meal (and state of glycogen stores before competition), and whether they consumed carbohydrate during the competition. In a crossover study, after an overnight fast, eight men who completed 6 sets of single-leg knee extensions at 70% of the 1RM, until 50% of full knee extension was no longer possible, depleted glycogen levels by approximately 29%. Within 6 hours of the completion of training, they restored up to 91% of their preexercise levels after consuming 1.5 gram of carbohydrate per kilogram body weight immediately after the resistance training session and again 1 hour later. However, when study subjects consumed water only immediately after training and again 1 hour later, they barely restored their muscle glycogen content to 75% of preexercise levels (104).

Net protein balance depends on both muscle protein synthesis and breakdown. And though carbohydrates have no effect on muscle protein synthesis, they help attenuate acute protein breakdown resulting from resistance exercise. The increase in protein breakdown likely depends on the resistance stimulus, the person's overall nutrition intake, and dietary intake before and during training or competition. One study in untrained young men found that acute protein breakdown increased by $51\% \pm 17\%$ after a bout of resistance training (16). In another study, the rate of both mixed muscle protein fractional

synthesis rate (muscle protein synthesis) and fractional breakdown rate (breakdown) were measured after a bout of either concentric or eccentric exercise in four untrained men and four untrained women. The rate of synthesis increased significantly above resting levels by 112% in the 3-hour period after exercise, 65% over the 24-hour period after exercise, and 34% over the 48-hour period after exercise. Muscle breakdown also increased by 31% and 18% in the 3- and 24-hour periods after exercise yet returned to baseline by 48 hours after exercise (110). Relatively small amounts of carbohydrate, somewhere between 30 and 100 grams, can sufficiently reduce muscle protein breakdown (18, 43). Although muscle protein breakdown has a role in overall protein balance, muscle protein synthesis plays a much bigger role in overall protein balance (43).

Supplementing with protein after a muscle-damaging bout of resistance training increases acute muscle protein synthesis. Varying amounts of protein have been used after resistance training to stimulate muscle protein synthesis, though it appears that 20 to 50 grams (providing about 8.5 g-20 g of essential amino acids) of a high-quality, high-leucine, fast protein (one that leads to a rapid rise in amino acids in the bloodstream) in younger individuals may maximally stimulate muscle protein synthesis, while 40 grams or more may be necessary in older adults (59, 109). Rice protein is an intermediate-speed leucine-poor protein (though the rise in leucine in the bloodstream appears quickly after consumption of rice protein); larger doses of rice protein effectively stimulate muscle protein synthesis to the same extent as a higher-quality protein when matched for leucine content (67). Therefore, the leucine content of the protein, and possibly the speed of leucine delivery, appears to be the determining factor for acute changes in maximal stimulation of muscle protein synthesis. A protein dose that contains 2 to 3 grams of leucine or 0.05 grams of leucine per kilogram body weight will effectively stimulate muscle protein synthesis in younger adults (96, 102, 136).

In addition to the acute increase in muscle protein synthesis, consistently supplementing with protein after resistance training led to small to moderate increases in muscle hypertrophy over time compared to conditions in which subjects were not given protein supplements (27, 122). Only one study to date has directly examined the relationship between acute measures of muscle protein synthesis and hypertrophy due to resistance training. In this study 23 young men who were recreationally active but had not participated in a resistance training exercise program in at least a year were tested for initial strength and resting and postexercise rates of muscle protein synthesis. Then they participated in a 16-week linear resistance training program consisting of two days of lower body (e.g., leg press, leg extension, leg curl, and calf press) and two days of upper body (e.g., chest press, shoulder press, seated row, lat

pulldown, biceps curl, triceps extension) exercises performed first thing in the morning after an overnight fast. After exercise each day and at breakfast on nontraining days they consumed a nutrition drink containing 30 grams of milk protein, 25.9 grams of carbohydrate, and 3.4 grams of fat. The study authors found no relationship between the acute rise (6 hours posttraining) in muscle protein synthesis due to exercise training and the nutrition drink provided and muscle hypertrophy. However, the authors also noted that changes in muscle protein synthesis with training were not uniform between study subjects; therefore, though acute changes in muscle protein synthesis are important for muscle hypertrophy, they are not the sole factor that predicts a person's potential to increase muscle growth (86). Thirty grams of milk protein, as used in this study, provides an estimated 24 grams of casein, 6 grams of whey protein, and a total of 2.8 grams of leucine (139).

Concurrent Training

Exercise interference is a concept suggesting that aerobic endurance exercise, when combined with resistance training (back-to-back sessions), blunts gains in strength compared to resistance training alone but results in improvements in aerobic endurance performance (148). While interest in concurrent training-based research has increased over time, nutrition recommendations for concurrent training are often based on isolated studies examining the impact of nutrition interventions on aerobic endurance or resistance training alone. The consumption of carbohydrate after aerobic endurance exercise and before lifting can help suppress skeletal muscle breakdown. Also, given the impact of protein intake after aerobic endurance exercise on muscle protein synthesis, as well as the research suggesting that protein intake during prolonged (>2 h) resistance training supports greater rates of muscle protein synthesis during exercise compared to a carbohydrate control, athletes should consider consuming protein after aerobic endurance exercise, before lifting, or during their lifting session if performing aerobic endurance exercise first (107).

Daily Protein Intake

While ingestion of protein around training is beneficial for athletes to aid in recovery and adaptation, this may have relatively little benefit if daily intake is sufficient. For this reason, protein consumption with each meal represents an ideal strategy to meet these needs and maintain a positive physiological environment for athletes. Resistance training has the ability to increase muscle sensitivity to amino acids for 24 to 48 hours after exercise whereas the anabolic effect of a meal lasts approximately 3 to 5 hours. For optimal muscle remodeling, experts suggest consuming at least 20 to 40 grams of protein per meal and

eating meals every 3 to 4 hours (81, 107) in order to facilitate meeting daily protein targets.

The amount of data on muscle protein synthesis in children is limited because invasive measurement procedures are required (17), and no studies have examined muscle protein synthesis after training and postworkout protein ingestion in healthy children. However, one thing is clear with regard to children: Due to size differences as well as the impact of insulin and calorie intake on protein synthesis, their per-meal protein intake targets will naturally be lower in order to meet their protein needs (79), despite potentially having higher relative needs. For more information about protein needs by sport, see [table 11.3](#).

Nutrition for Various Performance Goals

Nutrition for Aerobic Endurance Performance and Recovery

- Aerobic endurance athletes should aim to consume 8 to 10 grams of carbohydrate during their most demanding training phases or competition and 1.4 to 2.0 grams of protein per kilogram body weight per day throughout their training.
- Athletes who eat at least 4 hours before competition should include approximately 1 to 4 grams of carbohydrate per kilogram body weight and 0.15 to 0.25 grams of protein per kilogram body weight (131). If the precompetition meal is consumed 2 hours before exercise, athletes should aim for approximately 1 gram of carbohydrate per kilogram body weight (120) and 0.15 to 0.25 grams of easily digestible protein per kilogram body weight.
- Athletes should consume 28 to 144 grams of a mixture of multiple types of carbohydrate (e.g., sucrose, fructose, and glucose or maltodextrin) per hour during prolonged aerobic endurance exercise to extend time until exhaustion and improve performance (32, 34, 66, 137).
- During prolonged activity in hot weather, athletes should consume sport drinks containing 20 to 30 mEq of sodium (460-690 mg with chloride as the anion) per liter, 2 to 5 mEq of potassium (78-195 mg) per liter, and carbohydrate in a concentration of 5% to 10% (2).
- After exercise, approximately 1.2 to 2.0 grams of carbohydrate per kilogram body weight should be consumed within 30 minutes after stopping the exercise. The addition of protein appears to facilitate glycogen replenishment if carbohydrate intake is below 1.2 grams per kilogram. Additionally, some, though not all, studies show that the addition of protein to carbohydrate after a workout may attenuate muscle breakdown and soreness and increase muscle protein synthesis. The ideal amount of protein and the time period after exercise in which it should be consumed are not clear based on the current body of literature and may depend on whether the exercise was performed in the fed versus fasted state and on total calories consumed after a workout (44, 57, 85). However, as a general guideline, at least 20 to 40 grams of high-quality protein should also be consumed within a 2-hour time period after aerobic endurance exercise (sooner may be better, though the exact time period has yet to be elucidated in the research) (77).
- Glycogen stores should be replenished after exercise and before the next bout of training. A regular diet with sufficient carbohydrate intake can restore glycogen over the course of a 24-hour period. For faster glycogen synthesis, athletes should eat or drink a high-carbohydrate meal immediately after exercise and at regular intervals thereafter. This is especially important for athletes who train again less than 24 hours after their initial training session (61).

Nutrition for Strength

- Athletes should consider supplementing with carbohydrate before and during competition to maintain strength and minimize muscle breakdown (47). In general, strength or speed athletes should consume 5 to 6 grams of carbohydrate per kilogram body weight per day during peak training periods. As little as 30 grams of carbohydrate after training may reduce muscle breakdown. The ideal time period for carbohydrate consumption after resistance training has yet to be fully clarified; however, consuming carbohydrate sooner (rather than waiting) may be more beneficial, particularly if training was carried out in a fasted state (43).
- Because low glycogen can impair muscle force, athletes should consume higher-energy carbohydrates immediately following weightlifting or strength and power competitions if they must compete or train again over the course of the 24-hour period after their initial competition. Otherwise, they can consume enough carbohydrate over the course of the day to restore glycogen levels before the next bout of training or competition (47, 104).
- In general, strength or speed athletes should consume 1.4 to 2.0 grams of protein per kilogram body weight per day, even if their sport or training includes an aerobic endurance component. After resistance training, younger individuals should consume at least 20 to 50 grams (providing about 8.5 to 20 grams of essential amino acids) of a high-quality, high-leucine protein (2-3 g), while older adults should consume 40 grams or more to maximally stimulate muscle protein synthesis in the acute time period after training (109). If the exercise was performed in a fasted state (first thing in the morning or more than 3 hours after the last protein-rich meal), the protein should be consumed within 30 minutes after the end of the session; but if the exercise was performed in a fed state (preexercise protein-rich meal or supplement), this postexercise time window may be considerably longer (10).
- Between 30 and 100 grams of high-energy carbohydrates should be consumed after muscle-damaging exercise to reduce muscle protein breakdown (18, 43).
- Adult athletes should eat meals containing at least 20 to 30 grams of a higher-leucine protein.

Nutrition for Hypertrophy

- Ingestion of between 30 and 100 grams of energy-dense, moderate to high glycemic index carbohydrates may be consumed after muscle-damaging exercise to help reduce muscle protein breakdown (18, 43).
- After resistance training, younger individuals should consume at least 20 to 50 grams (providing about 8.5 to 20 grams of essential amino acids) of a high-quality, high-leucine protein (>2-3 g), while older adults should consume 40 grams or more to maximally stimulate muscle protein synthesis in the acute time period after training (109).
- Adult athletes should eat meals containing at least 20 to 40 grams of a higher-leucine protein every 3 to 4 hours.

Nutrition for Muscular Endurance

- Maintain adequate hydration by preventing water weight losses exceeding 2% of body weight.
- During prolonged training or competition, consider a carbohydrate-electrolyte beverage to delay fatigue and improve performance, particularly if performing after an overnight fast (6, 142).
- Fully replace glycogen stores before the next training session or competition.
- Consume protein after training or games to minimize muscle damage and soreness (42).

TABLE 11.3 Sport-Specific Protein Needs

Sport	Daily protein needs in grams per kilogram body weight	Sample daily protein intake	Postexercise protein needs	Postexercise protein examples
Low to moderate aerobic endurance activity (e.g., jogging, triathlon)	1.4-1.8	150 lb (68 kg) individual: • 1 cup (240 g) higher-protein Greek yogurt (18-22 g protein) • Grilled chicken sandwich (28 g protein) • Spaghetti with 4 oz (113 g) meatballs (20 g protein) • Salmon (6 oz [170 g]) (34 g protein)	0.2-0.5 g per kilogram body weight	• 1 higher-protein nutrition bar (20 g protein) • 1 cup (240 g) higher-protein Greek yogurt (18-22 g protein)
Elite aerobic endurance athlete or one who trains intensely	>1.6	140 lb (64 kg) individual: • 1 cup (240 g) higher-protein Greek yogurt (18-22 g protein) • Grilled chicken sandwich (28 g protein) • Spaghetti with 6 oz (170 g) meatballs (30 g protein) • 95% lean beef (4 oz [113 g]) (30 g protein) • 1/2 cup low-fat cottage cheese (12 g protein)	0.2-0.5 g per kilogram body weight	• 20 fl oz (591 mL) 1% milk or higher-protein soy milk (20 g protein) • 1 scoop whey protein (20-25 g protein)
American football	1.6-2.2, though likely at the upper end of this range for those playing at an intense level	240 lb (109 kg) individual: • 5 large eggs (30 g protein) • Roast beef sandwich (28 g protein) • 8 oz (227 g) grilled salmon	Not clarified by research; therefore, athletes may want to follow the guidelines for resistance training and consume a minimum of 30 g of a fast, high-quality, leucine-rich (2-3 g leucine) protein	• Tofu and berry shake (1/2 cup [125 g] tofu, 1 scoop soy protein powder, 1/2 cup [75 g] mixed berries) (23 g protein)

Sport	Daily protein needs in grams per kilogram body weight	Sample daily protein intake	Postexercise protein needs	Postexercise protein examples
		(45 g protein) • 6 oz (170 g) chicken thighs with 1 cup (185 g) cooked quinoa (50 g protein) • 1 cup (240 g) higher-protein Greek yogurt (18-22 g protein) • 12 fl oz (355 mL) 1% milk with 1 scoop whey protein (34 g protein)		• 1 scoop whey protein with 4 tbsp (32 g) powdered peanut butter (38 g protein)
Gymnastics	Dietary restriction increases protein needs up to 2.2 (if they restrict calories); no studies conducted on gymnasts' protein needs	105 lb (48 kg) individual: • 3 large egg omelet with 1 oz (28 g) cheese (25 g protein) • 5.3 oz (150 g) Greek yogurt with 1 oz (28 g) almonds and 1/2 cup (75 g) berries (18 g protein) • Chicken shawarma wrap (4 oz [113 g] grilled chicken, 1 whole wheat pita, 2 tbsp [30 g] hummus [37 g protein]) • 3 oz (85 g) shrimp with 1/2 cup (85 g) black beans (29 g protein)	0.2-0.5 g per kilogram body weight	• 1 cup (210 g) low-fat cottage cheese with 1/2 cup (75 g) pineapple chunks (25 g protein) • 2 scoops (60 g) plant-based protein powder, 1 cup (237 mL) almond milk, 1 tbsp (15 g) chia seeds (28 g protein)
Team sports	Likely 1.6-2.0 g, depending on the sport and	165 lb (75 kg) individual: Follow the sample daily meal plan for	~0.2-0.5 g per kg per hour (minimum of 20 g)	• 1 scoop of whey protein in 8 fl oz (237 mL) skim

Sport	Daily protein needs in grams per kilogram body weight	Sample daily protein intake	Postexercise protein needs	Postexercise protein examples
	intensity of exercise	the elite aerobic endurance athlete above and add 5 oz (142 g) chicken thighs with 1 cup (185 g) cooked quinoa (43 g protein)		milk (30 g protein)
Weightlifting	1.6-2.0	212 lb (96 kg) individual: 4 large eggs omelet with 1 oz (28 g) cheese (30 g protein) 6 oz (170 g) grilled salmon with 1 cup (185 g) cooked quinoa (42 g protein) 6 oz (170 g) sirloin steak (46 g protein) 6 oz (170 g) shrimp, 1/2 cup (85 g) black beans, 2 soft taco size whole grain tortillas (58 g protein)	Studies have used 20-70 g protein in close temporal proximity to resistance exercise, though a protein containing approximately 2-3 g leucine per serving of protein is recommended	1 scoop egg or whey protein (or 2 scoops plant-based protein) with 4 tbsp (32 g) powdered peanut butter (38 g protein)
Wrestling	2.2, though dietary restriction increases protein needs	190 lb (86 kg) individual: Follow the sample daily meal plan for the weightlifter, but add the following: 1 cup (210 g) low-fat cottage cheese, 1/2 cup (75 g) pineapple chunks (25 g protein)	0.2-0.5 g per kilogram (minimum of 20 g of high-quality protein)	1 scoop of whey protein in 8 fl oz (237 mL) skim milk (28 g protein) or 2 scoops of whey protein in water (30-44 g protein)

Nutrition Strategies for Altering Body Composition

Athletes who want to alter their body composition typically need to gain muscle, lose body fat, or both. Although there are some scenarios (typically in those with eating disorders) in which an athlete may need to gain both muscle and fat, this section focuses on gaining muscle and losing body fat.

The first step toward altering body composition involves estimating calorie needs (though **calorie** is the common term, the technical term is **kilocalorie**). The number of calories an athlete needs each day depends on a number of factors including genetics, body weight, body composition, training program, and age. Children and adolescents also need calories for growth and development.

Basal metabolic rate (BMR) is by far the largest contributor to total energy expenditure, accounting for approximately 65% to 70% of daily energy expenditure (63, 114). It is a measure of the calories required for maintaining normal body functions such as respiration, blood circulation, and gastrointestinal and renal processing. Basal metabolic rate and **resting metabolic rate (RMR)** are often used interchangeably, though they are slightly different. Basal metabolic rate is measured after an overnight fast (12-14 h without food) with the subject resting supine and motionless but awake (2, 40). Resting metabolic rate is often used instead of BMR due to the ease of measurement (an overnight fast is not required), yet it is 10% to 20% higher than BMR due to increased energy expenditure resulting from recent food intake or physical activity completed earlier in the day. Several factors influence BMR and RMR, particularly fat-free mass, which explains approximately 70% to 80% of the difference in RMR among individuals (93, 115, 138). Other factors include age, nutrition status, genetics, and differences in endocrine functioning (such as hypo- or hyperthyroidism).

The second-largest component of an individual's energy requirement is the energy expended in physical activity. Of all the components, it is the most variable among individuals. The number of calories expended through physical activity increases with the frequency, intensity, and duration of the training program, as well as nontraining daily activity (e.g., washing a car or doing housework). Typically, 20% to 30% of total daily energy expenditure is from physical activity, though this figure may be considerably higher in athletes (63, 114). The highest energy costs are seen in aerobic activities performed by large athletes for long periods of time, whereas the lowest energy costs are associated with skill and power sports performed by smaller athletes.

The thermic effect of food, also known as **diet-induced thermogenesis**, is the increase in energy expenditure above the RMR that can be measured for several hours following a meal. The thermic effect of food includes the energy cost of digestion, absorption, metabolism, and storage of food in the body. The thermic effect of food accounts for approximately 10% to 15% of total calories burned each day (63, 114).

Many different equations can be used to calculate calorie needs, including the Cunningham equation and equations by the Institute of Medicine (IOM), which are prediction equations. The various IOM equations take sex, body weight, height, age, and physical activity levels into account to predict the estimated energy requirement (EER) (138). RMR can then be multiplied by an activity factor from 1.2 (sedentary) to 1.9 (heavy physical activity) to predict energy requirements (50). However, these equations do not take into account how muscle mass affects RMR (90). The Cunningham equation takes fat-free mass (FFM) into account to estimate RMR, making it more applicable to athletes (135).

$$\text{RMR} = 500 + 22 (\text{FFM in kg})$$

After using the Cunningham equation to estimate RMR based on FFM, an activity factor can be used to estimate total daily energy expenditure. Instead of using a basic activity factor, the strength and conditioning professional may choose to use metabolic equivalents (MET values), an estimate of caloric expenditure during activity. One MET is the energy equivalent of sitting quietly; therefore, the greater the intensity of exercise, the higher the MET value (5).

Another, more laborious method that can be used by very motivated athletes is to record dietary intake for a minimum of three consecutive, representative days during a period of stable body weight. The individual's daily energy requirement is assumed to equal the average number of calories consumed each day. The pitfall of this method is that recording food intake usually alters normal eating habits and people do not always accurately record their food intake (55, 121). Finally, a very simple method for quickly assessing calorie needs is presented in [table 11.4](#) (24).

Weight Gain

An athlete's ability to gain weight depends on numerous factors; the ones that can be controlled include diet and training. The off-season should be used as a time to make dietary changes that focus on weight gain since athletes do not have the pressures of competition on their mind.

If athletes increase their calorie intake dramatically and consistently, they could gain more fat than they would like. A general guideline, which should be adjusted based on the athlete, is to consume approximately 500 additional calories per day in order to gain weight (113). Eating larger portion sizes,

increasing meal frequency, and focusing on choosing calorie-dense foods are all strategies that can help an athlete gain weight. In addition to increasing total calorie intake, athletes should ensure that they are eating enough protein to maximize gains in lean body mass: 1.6 to 2.2 grams per kilogram body weight per day. Increasing an athlete's protein intake may make weight gain tough, given the profound effect that protein has on satiety, as well as the increased thermic effect of feeding associated with higher-protein diets. However, overfeeding protein is advantageous if the athlete can sustain the diet. In a randomized, controlled overfeeding study, 16 healthy adults lived in a metabolic ward for eight weeks and were overfed a diet containing low (5% of total calories), normal (15% of total calories), or high (25% of total calories) protein. The subjects who consumed the normal- and high-protein diet stored approximately 45% of the excess calories as lean body mass, whereas those on the low-protein diet stored 95% of the excess calories consumed as body fat (20). In addition to making dietary changes, athletes who want to gain weight should consider supplementing with creatine monohydrate, a supplement that safely and effectively increases lean body weight (73).

TABLE 11.4 Physical Activity Level Values and Activity Coefficients

Physical activity level (PAL)*	Examples	Physical activity coefficient (males/females)
Sedentary (≥ 1.0 to < 1.4)	Common activities of daily living (ADLs)	1.00/1.00
Low active (≥ 1.4 to < 1.6)	ADLs and 30-60 min of daily moderate activity	1.11/1.12
Active (≥ 1.6 to < 1.9)	ADLs and ≥ 1 h of daily moderate activity	1.25/1.27
Very active (≥ 1.9 to < 2.5)	ADLs and ≥ 1 h of daily moderate activity and 1 more h of vigorous activity or 2 h of moderate activity	1.48/1.45

*PAL = ratio of energy expenditure to basal daily energy expenditure ("1.0" means total energy expenditure equals basal daily energy expenditure).

Reprinted by permission from L.M. Burke and E.S. Rawson, "Fueling and Nutrition," in *NSCA's Essentials of Sport Science*, edited for the National Strength and Conditioning Association by D.N. French and L. Torres Ronda (Human Kinetics, 2022), 366.

Finally, though athletes may know their energy and protein needs, putting this knowledge into practice can be perplexing. Therefore, regular nutrition counseling (or coaching) by a sports dietitian or sports nutritionist with an advanced degree is a recommended strategy for facilitating greater weight gain.

A study of 21 elite athletes with heavy training loads, who were randomized to receive nutrition counseling or to eat ad libitum, showed that over the course of the 8- to 12-week weight gain period, the group receiving dietary counseling had greater total and lean body mass gains during the intervention and after 12 months; this means that the group receiving nutrition counseling continued to benefit from the nutrition guidance for several months after counseling ended (41).

Weight (Fat) Loss

Athletes participating in a variety of sports may need to lose body fat to improve aspects of performance such as speed or aerobic endurance. Plus, regulating body weight may give certain athletes a mental advantage going into competition (108). Additionally, maintaining or making weight is a recurring aspect of sports that include weight classes, weight limitations, or judging based on appearance such as weightlifting, wrestling, boxing, lightweight crew, and gymnastics.

Despite diet books on the bestseller lists, there is no ideal diet that works for everyone. Instead, studies show that a variety of types of diets, including low-carbohydrate and low-fat diets, result in weight loss as long as the people following them are consuming fewer calories than needed to maintain weight. Additionally, there appears to be no difference between the amount of weight loss on a low-carbohydrate as compared to a low-fat (and therefore higher-carbohydrate) diet (19, 39). Total calorie intake and dietary adherence, the ability to stick with a diet over time, are the two most important factors that predict successful weight loss. However, a good portion of weight loss during dieting may come from muscle (144). Further, because muscle protein synthesis is an energy-expensive process, during times of dieting caloric restriction may decrease muscle protein synthesis. Athletes who want to maintain muscle and lose body fat while dieting should consume about 1.8 to 2.7 grams protein per kilogram body weight per day (or approximately 2.3-3.1 g protein per kilogram fat-free mass per day) in addition to maintaining a moderate energy deficit of approximately 500 calories per day (91).

There is no one ideal diet. Instead, athletes need to choose a dietary approach based on whether it is safe for them, contains enough protein to meet their needs, and fits their lifestyle so that they can easily adhere to it.

To be sustainable in the long term, diets should be individualized; be easy to adhere to; and take into account lifestyle habits, medical history (including

diabetes, insulin resistance, other diseases, and medical concerns), diet history, and food preferences while providing all of the nutrients an athlete needs to train and perform optimally. Additionally, research shows that ongoing behavior therapy and support can improve long-lasting results (80).

Overweight and Obesity

Overweight and **obesity**, both of which reflect an accumulation of excess body fat, increase a person's risk of morbidity from hypertension; dyslipidemia; coronary artery disease; gallbladder disease; stroke; type 2 diabetes; sleep apnea; osteoarthritis; respiratory problems; and endometrial, breast, prostate, and colon cancers (92). Obesity is categorized as a disease and affects 34.9% of adults and 17% of children in the United States (96). At a population level, overweight and obesity are typically classified based on a **body mass index (BMI)** of 25 to 29.9 kg/m² and ≥30 kg/m², respectively ([table 11.5](#)). However, this evaluative approach may have very little applicability to athletes. In reality, BMI is a measure of excess weight as opposed to excess body fat (92) because it cannot distinguish between excess fat and muscle or bone mass (26). Therefore, BMI can overestimate body fat in athletes and others with muscular builds and underestimate body fat in older persons or those who have lost muscle mass (95). Finally, age, sex, ethnicity, and muscle mass affect the association between BMI and body fat. Strength and conditioning professionals should use other more accurate measures of body composition such as skinfolds, air displacement plethysmography (ADP), multifrequency bioelectrical impedance analysis, or **dual-energy X-ray absorptiometry (DXA)** in athletes with more muscle than their age- and sex-matched counterparts (74).

The causes of obesity are complex and include an interaction of genes and environment, involving social, behavioral, cultural, metabolic, physiological, and genetic factors. However, many treatment options are effective, including dietary therapy, altering physical activity patterns, behavior therapy techniques, pharmacotherapy, and surgery. Some patients and clinicians employ a combination of these treatments. The initial goal for weight loss in overweight and obese individuals should be 10% of initial weight within six months (92). Additionally, **target body weight** can be calculated using body fat percentage and FFM (i.e., target body weight = fat free mass/[100 percent desired percent body fat]). Where desired, percent body fat is based on measurement type (e.g., ADP, DXA, skinfolds) and team, sport, and position-specific normative body composition data.

Low-Carbohydrate Diets

The popularity and effectiveness of low-carbohydrate diets may have less to do with cutting carbohydrates (in healthy individuals) and more to do with a temporary decrease in water weight, cutting calories, and increasing protein intake. When people significantly reduce their carbohydrate intake, they deplete their glycogen stores (carbohydrate stored with three to four parts water) and therefore quickly drop pounds from water weight. Once carbohydrates are reintroduced into the diet, they can rapidly gain the weight back (71).

Over time, however, low-carbohydrate diets may work for some people because of the consumption of increased levels of protein. Protein increases feelings of fullness (satiety) in a dose-dependent manner—the more protein consumed at one sitting, the greater its effect on satiety. However, at this time the “optimal dose” for maximum satiety remains unclear. Protein also increases the thermic effect of feeding. More calories are burned during digesting and processing protein than carbohydrate or fat. And finally, protein helps spare metabolically active lean muscle tissue during weight loss. This is important because muscle burns a few more calories at rest than fat does, and over time this can affect body weight. Plus, more muscle may mean that a person can train harder and therefore burn more calories during training (91).

Though cutting carbohydrates may be detrimental for a number of competitive athletes, particularly during precompetition and competition phases, it is a very effective approach for someone with insulin resistance (a condition that leads to the buildup of glucose in the blood because the body does not use insulin effectively) as well as type 2 diabetes. It may also be an effective approach for overweight and obese individuals (51, 54, 149).

No single body fat measure should be used to assess health, disease, or disease risk (26). Overweight and obesity combined with other risk factors, including high blood pressure, high low-density lipoprotein cholesterol, low high-density lipoprotein cholesterol, high triglycerides, high blood glucose, physical inactivity, family history of premature heart disease, or cigarette smoking, increase a person’s risk for heart disease (92).

Although all obese individuals share the trait of excess body fat, they cannot be treated homogeneously. They must be screened for coexisting illnesses such as diabetes, orthopedic problems, cardiac disease, psychological disorders such as **binge-eating disorder** or depression, social and cultural influences, and readiness for change. Obese athletes who are mandated to lose weight (e.g., by a physician) pose a special challenge because losing weight is dictated by an external source and is not an internalized goal. Weight loss takes a great deal of dedication from within the individual, and these athletes may need to work closely with a mental health professional or registered dietitian to help them meet their goal.

TABLE 11.5 Classification of Adult Weight by Body Mass Index (BMI) and Associated Disease Risk

			DISEASE RISK ^b RELATIVE TO NORMAL WEIGHT AND WAIST CIRCUMFERENCE	
Classification	Obesity class	BMI ^a (kg/m ²)	Men <102 cm (<40 in) Women <88 cm (<35 in)	Men >102 cm (>40 in) Women >88 cm (>35 in)
Severely underweight		<16.5		
Underweight		<18.5		
Normal		18.5-24.9		
Overweight		25.0-29.9	Increased	High
Obesity	I	30.0-34.9	High	Very high
	II	35.0-39.9	Very high	Very high
Extreme obesity	III	≥40	Extremely high	Extremely high

Note: Increased waist circumference can also be a marker for increased risk even in persons of normal weight.

^aBMI may overestimate fat in athletes and others with muscular builds and underestimate fat in older persons and others who have lost muscle. To estimate BMI using kilograms and meters, use this equation: body weight (kilograms) / height (meters)². To estimate BMI using pounds and inches, use this equation: body weight (pounds) / height (inches)² × 703.

^bDisease risk for type 2 diabetes, hypertension, and cardiovascular disease.

Reprinted from NHLBI Obesity Education Initiative Expert Panel on the Identification, Evaluation, and Treatment of Obesity in Adults (1998).

Rapid Weight Loss

Though there is no uniform definition in the literature, *rapid weight loss* generally refers to quick weight loss, faster than can be achieved by lowering calorie intake and increasing exercise, in a short period of time (38). Athletes may use a variety of techniques to cut weight quickly to compete in a desired weight class, meet a weight goal set by their coach, or improve performance. Potentially dangerous weight loss techniques may include fasting, fad diets, **voluntary dehydration** (diuretics, sauna, water and salt manipulation, wearing multiple layers of clothing), excessive spitting, self-induced vomiting, laxative abuse, and inappropriate or excessive use of thermogenic aids (38).

Athletes who attempt to lose too much weight too quickly may lose lean body mass, feel fatigued, experience headaches or mood swings, put their training and performance in jeopardy, and suffer from several potentially serious side effects. These include dehydration, heat illness, muscle cramping, fatigue, dizziness, suppressed immune system functioning, hormone imbalances, hyperthermia, reduced muscle strength, decreased plasma and blood volume,

low blood pressure, electrolyte imbalances, kidney failure (diuretic abuse), fainting, and death (in extreme cases) (38).

The strength and conditioning professional should be able to recognize the signs and symptoms associated with rapid weight loss techniques and refer athletes to the appropriate professional while also communicating concerns to the rest of the coaching staff. In addition, the strength and conditioning professional may want to consider documenting the steps he or she has taken to assist the athlete (for good record keeping and to prevent liability) while also working with a medical doctor or registered dietitian to set appropriate weight goals after body composition, dieting, medical, and eating disorder history have been taken into account. Some athletes may need to reconsider which weight class they are competing in if they cannot make weight without putting their health or performance in jeopardy.

Relative Energy Deficiency in Sport (RED-S)

Relative energy deficiency in sport (RED-S) encompasses various physiological dysfunctions resulting from low energy availability (LEA), where an athlete's energy intake does not match the energy expended during exercise. This mismatch affects metabolic rate, menstrual function, bone health, immunity, protein synthesis, and cardiovascular health, among others. This phenomenon was initially recognized in 2014 with the international RED-S consensus statement (88) and has continued to encompass additional research. For more details on the implications of RED-S, see these previous papers (25, 89).

LEA is quantified by **energy availability (EA)**, calculated as energy intake minus exercise energy expenditure, relative to fat-free mass (FFM). Optimal EA for healthy function is around 45 kilocalories per kilogram FFM per day, while severe perturbations can occur below 30 kilocalories per kilogram FFM per day (25, 88). LEA adversely affects various systems, including the endocrine system, leading to disruptions in hormonal balance, particularly in female athletes, and possibly in males, though male athletes appear to be more physiologically resilient at lower EA values. LEA may be one of the strongest indicators of injury risk for athletes (133), with oligomenorrheic and functional hypothalamic amenorrheic (FHA) athletes reporting higher percentages of severe musculoskeletal injuries resulting in greater durations of time lost from sport (134).

Menstrual dysfunction, including **amenorrhea**, is a common consequence of sustained LEA in female athletes, due to the impact on the hypothalamic–pituitary–gonadal (HPG) axis. Chronic LEA also compromises bone health, leading to decreased bone mineral density and increased risk of bone stress injuries, particularly in sports with high energy demands. Metabolic changes, including decreased RMR, and hematological alterations, such as iron deficiency, are also associated with LEA. Growth and development, cardiovascular function, gastrointestinal health, and immune function are also affected, particularly at very low levels of LEA and over longer periods of time. Psychological implications include increased risk of depression, eating disorders, and impaired stress management.

Performance-wise, LEA can impair recovery, muscle function, and glycogen storage, affecting physical and psychological capacity, training quality, and injury risk (89). However, despite these associations, direct evidence of LEA's impact on sport performance is limited. Studies on elite athletes suggest that LEA may impair sport performance indirectly by acutely affecting fuel storage and use (132), reducing training volume, and increasing risk of illness (35, 36). Yet further research with robust protocols is needed to fully understand the acute

and chronic effects of LEA on athletic performance. Recognizing and addressing LEA is crucial for optimizing athlete health, performance, and long-term well-being. Health care practitioners and coaches should be vigilant for signs of LEA and its associated health implications, implementing strategies to mitigate risks and support athletes in achieving optimal energy balance and performance outcomes.

Feeding and Eating Disorders

Eating disorders, including binge-eating disorder, **anorexia nervosa**, and **bulimia nervosa**, are serious mental health disorders that can affect both men and women, appear at any point in life, and increase risk of mortality. Those with eating disorders have a high prevalence of other psychiatric disorders, including anxiety disorders, mood disorders such as depression, impulse-control disorders, and substance abuse disorders (49, 58).

Studies show an increased prevalence of both **disordered eating** and **eating disorders** in athletes compared to controls (45, 128, 151). In particular, athletes in sports with weight classes such as wrestling, those in sports that emphasize leanness such as cross country running, and aesthetic sports such as gymnastics may be more prone to disordered eating and eating disorders (45, 128, 151). Signs of disordered eating may include restrictive eating; fasting; skipping meals; and taking diet pills, laxatives, or diuretics. However, those with disordered eating do not meet the full diagnostic criteria for an eating disorder (45).

Eating disorders are multifactorial diseases and, as such, require a multidisciplinary team approach. The strength and conditioning professional must be aware of signs and symptoms of an eating disorder and have a referral network in place so athletes can get the psychological, medical, and nutrition help they need from experts who work with eating disorders.

Anorexia Nervosa

Anorexia nervosa is characterized by a distorted body image and an intense fear of gaining weight or becoming fat, leading those with this disease to excessive calorie restriction and severe weight loss (8). Those with anorexia nervosa also put a great deal of emphasis on their weight or shape yet do not recognize the seriousness of their illness. Also, people with anorexia nervosa typically engage in ritualistic behaviors including repeated weighing, cutting food into small pieces, and carefully portioning their food (92).

Two subtypes fall under anorexia nervosa. The restricting type does not regularly binge eat or purge, while the binge-eating or purging type regularly engages in binge eating or purging.

The incidence peaks in the early 20s for both males and females, and the lifetime prevalence in females ages 10 to 39 years ranges from 0.21% to 0.89%, and in males ages 10 to 39 years it ranges from 0.19% to 0.67% (143). Estimates based on data modeling reveal that 14% of females with anorexia nervosa and 28.2% of males seek treatment (143). It is important to note that among all mental health disorders, anorexia nervosa has the highest mortality

rate (49, 143). For more information about anorexia nervosa, the reader is encouraged to read the American Psychiatric Association's *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-V)* (8).

Symptoms of Anorexia Nervosa

Signs and Symptoms of Anorexia Nervosa

- Extremely restricted eating
- Intense and excessive exercise
- Extreme thinness (emaciation)
- A relentless pursuit of thinness and unwillingness to maintain a healthy weight
- Intense fear of gaining weight
- Distorted body image or self-image
- Denial of the seriousness of low body weight

Serious Health Consequences of Anorexia Nervosa

- Thinning of the bones (osteopenia or osteoporosis)
- Mild anemia
- Muscle wasting and weakness
- Delayed puberty
- Severe constipation
- Low blood pressure
- Slowed breathing and pulse
- Damage to the structure and function of the heart
- Feeling tired all the time
- Infertility
- Brain damage
- Multiple organ failure

Reprinted from National Institute of Mental Health, "Eating Disorders: What You Need to Know," U.S. Department of Health and Human Services, National Institutes of Health, retrieved August 7, 2025, <https://www.nimh.nih.gov/health/publications/eating-disorders>.

Binge-Eating Disorder

Previously categorized under "eating disorder not otherwise specified," binge-eating disorder has its own category in the *DSM-V* (8). Binge-eating disorder is characterized by repeated episodes, occurring at least once a week for a period of three weeks, of uncontrolled binge eating (eating significantly more food in a short period of time than most people would eat under the same circumstances) (55). The binge-eating episodes are associated with three or more of the following:

- Eating much more rapidly than normal

- Eating until feeling uncomfortably full
- Eating large amounts of food when not feeling physically hungry
- Eating alone because of feeling embarrassed by how much one is eating
- Feeling disgusted with oneself, depressed, or very guilty afterward (55)

Because binge-eating episodes are not followed by purging, as is the case with bulimia nervosa, people with binge-eating disorder are often overweight or obese. The lifetime prevalence for males ages 10 to 39 years is estimated to be 0.69% to 1.93% and 1.08% to 3.14% in females of the same age, with peak onset in the early to mid-20s (143). A lifetime prevalence of binge-eating disorder is associated with morbid obesity (BMI >40). Over the course of a lifetime, it is estimated that approximately 30% of males and 29.7% of females with binge-eating disorder receive treatment (143). Binge eating is associated with significant physical and psychological problems. Additionally, people with binge-eating disorder may feel embarrassed, guilty, or disgusted with their behavior and may attempt to hide their behavior by eating alone (55).

Bulimia Nervosa

Bulimia nervosa is characterized by recurrent consumption of food in amounts significantly greater than would customarily be consumed in a discrete period of time—for example, an entire pizza and a half-gallon of ice cream, plus a package of cookies. Purging follows episodes of binge eating and may include one or more of the following: self-induced vomiting, intense exercise, laxative use, or diuretic use. The bingeing and purging occur at least once a week for a period of three months (55). People with bulimia nervosa feel a lack of control over their eating during binge episodes and are more likely to be normal weight as opposed to underweight, are unhappy with their weight and body, and fear weight gain. Average age of onset for bulimia nervosa is in the early 20s, and the lifetime prevalence is 0.08% to 0.30% in males ages 10 to 39 and 0.56% to 1.7% in females of the same age range. Approximately 15.7% of males and 17.7% of females are estimated to receive treatment (143). For more information about bulimia, the reader is encouraged to refer to *DSM-V* (84).

Avoidant/Restrictive Food Intake Disorder

Avoidant/restrictive food intake disorder (ARFID) is an eating or feeding disturbance, including apparent lack of interest in eating or food; avoidance based on the sensory characteristics of food; or concern about aversive consequences of eating. This disorder is manifested by persistent failure to meet appropriate nutritional or energy needs associated with one (or more) of the following (55):

- Significant weight loss (or failure to achieve expected weight gain or faltering growth in children)
- Significant nutritional deficiency
- Dependence on enteral feeding or oral nutritional supplements
- Marked interference with psychosocial functioning

The disturbance is not better explained by lack of available food or by associated culturally sanctioned practice.

The eating disturbance does not occur exclusively during the course of anorexia nervosa or bulimia nervosa, and there is no evidence of a disturbance in the way in which one's body weight or shape is experienced.

The eating disturbance is not attributable to a concurrent medical condition or not better explained by another mental disorder. When the eating disturbance occurs in the context of another condition or disorder, the severity of the eating disturbance exceeds that routinely associated with the condition or disorder and warrants additional clinical attention (55).

Pica

People with pica eat nonnutritive substances for a period of at least one month. Common nonnutritive substances include clay, laundry starch, ice, cigarette butts, hair, or chalk (68). Those with pica may have electrolyte and metabolic disorders, intestinal obstruction, wearing away of tooth enamel, and gastrointestinal problems, among other issues. Testing for anemia is recommended since pica is associated with iron deficiency (68).

Rumination Disorder

Rumination involves chewing, reswallowing, or spitting of regurgitated food. To be classified as having this disorder, one must display this behavior, unrelated to any medical condition, for at least one month. Rumination disorder can occur alongside other eating issues or disorders (55).

Symptoms of Bulimia Nervosa

Signs and Symptoms of Bulimia Nervosa

Vomiting
Taking laxatives
Exercising excessively
Fasting

Serious Health Consequences of Bulimia Nervosa

Chronically inflamed and sore throat

Swollen salivary glands in the neck and jaw area
Worn tooth enamel, increasingly sensitive and decaying teeth
Acid reflux disorder and other gastrointestinal problems
Intestinal distress and irritation
Severe dehydration
Electrolyte imbalance

Reprinted from National Institute of Mental Health, "Eating Disorders: What You Need to Know," U.S. Department of Health and Human Services, National Institutes of Health, retrieved August 7, 2025, <https://www.nimh.nih.gov/health/publications/eating-disorders>.

Eating Disorders: Management and Care

It is not the responsibility of the strength and conditioning professional to treat or diagnose an eating disorder. It is his or her ethical responsibility to assist the athlete in attaining the proper diagnosis from a qualified physician and treatment from a qualified treatment team. Therefore, strength and conditioning professionals should be mindful of the symptoms of each eating disorder as well as signs of disordered eating. Keep in mind that abnormal eating patterns and amenorrhea alone are not indicative of an eating disorder. A professional who is experienced and qualified in diagnosing and treating eating disorders should be contacted when an athlete's behavior is concerning.

Eating Disorder Resources

National Eating Disorders Association

www.nationaleatingdisorders.org

International Association of Eating Disorders Professionals

www.iaedp.com

The Renfrew Center Foundation

www.renfrewcenter.com

National Association of Anorexia Nervosa and Associated Disorders

www.anad.org

The Meadows Ranch

www.meadowsranch.com

The strength and conditioning professional is not responsible for treating eating disorders but instead should be aware of the symptoms associated with an eating disorder and refer athletes to the appropriate professional.

Conclusion

The primary role of nutrition in strength and conditioning is to support athletic performance. A general understanding of nutrition principles and applications is essential for strength and conditioning professionals so that they can provide consistent, accurate information to their athletes while also being able to identify the potential signs and symptoms of an eating disorder. And though pre-, during-, and postcompetition nutrition can lead to better performance, a nutritionally sound daily diet should be emphasized as well for overall health, greater training adaptations, and performance.

KEY TERMS

amenorrhea
anabolic window
anorexia nervosa
basal metabolic rate (BMR)
binge-eating disorder
body mass index (BMI)
bulimia nervosa
calorie
carbohydrate-electrolyte (carbohydrate) drink
carbohydrate loading
creatine kinase
diet-induced thermogenesis
disordered eating
dual-energy X-ray absorptiometry (DXA)
eating disorders
energy availability (EA)
exogenous
isocaloric
kilocalorie
obesity
net protein balance
precompetition meal
relative energy deficiency in sport (RED-S)
resting metabolic rate (RMR)
target body weight
voluntary dehydration

STUDY QUESTIONS

1. Which of the following is a key strategy for precompetition nutrition in athletes?
 - a. consuming high-fat and high-fiber foods to promote satiety
 - b. choosing carbohydrates and fluid to maximize energy and hydration
 - c. avoiding protein intake before competition to prevent muscle soreness
 - d. trying new foods during practice sessions to minimize stomach discomfort
2. Which of the following statements about nutrition during aerobic endurance events and intermittent-activity sports is accurate?
 - a. Protein intake has no impact on performance or muscle damage.
 - b. Athletes should avoid consuming carbohydrates during prolonged exercise to prevent stomach discomfort.
 - c. Proper hydration and carbohydrate intake are essential to maintain performance and prevent dehydration.
 - d. Sport drinks with a carbohydrate concentration higher than 10% are ideal for quick absorption.
3. What are the recommended amounts of carbohydrate and protein intake for aerobic endurance athletes during their most demanding training phases or competition?
 - a. carbohydrate: 1.5 to 2.0 grams per kilogram body weight per day; protein: 1.40 to 2.0 grams per kilogram body weight per day
 - b. carbohydrate: 8 to 10 grams per kilogram body weight per day; protein: 1.40 to 2.0 grams per kilogram body weight per day
 - c. carbohydrate: 1.5 to 2.0 grams per kilogram body weight within 30 minutes after exercise; protein: 10 to 40 grams within a 32-hour time period after aerobic endurance exercise
 - d. carbohydrate: 28 to 140 grams per hour during prolonged aerobic endurance exercise; protein: 1.5 to 2.0 grams per kilogram body weight within 30 minutes after exercise
4. What are the recommended macronutrient intakes for strength or speed athletes during peak training periods?
 - a. carbohydrate: 1.4 to 1.7 grams per kilogram body weight per day; protein: 5 to 6 grams per kilogram body weight per day
 - b. carbohydrate: 30 to 100 grams immediately following weightlifting or strength competitions; protein: 20 to 30 grams of high-leucine protein every 3 to 4 hours
 - c. carbohydrate: 5 to 6 grams per kilogram body weight per day; protein: 1.4 to 1.8 grams per kilogram body weight per day